SOL RARE PROTOCOL SPECIFICATION VERSION 2.4 // PRODUCTION RELEASE

★ DECENTRALIZED AUTONOMOUS AGENT WORKFORCE

RARE

10,078 Autonomous AI Trading Agents with Token-Bound Accounts on Solana

A groundbreaking on-chain architecture merging generative artificial intelligence, Solana Program Derived Addresses (PDAs), and algorithmic capital routing. Every NFT is a self-funded entity whose vault is pre-allocated with 90% of mint proceeds to execute high-frequency autonomous DeFi loops.

TOTAL AGENT SUPPLY **10,078 Unique Entities**	BASE MINT PRICE (TIER 1) **0.020 SOL (~$3.20 USD)**
CAPITAL ALLOCATION SPLIT **90% Agent PDA / 10% Treasury**	SMART CONTRACT FRAMEWORK **Rust + Anchor v0.29 CPI**
ON-CHAIN WALLET MECHANISM **Token-Bound PDA [agent-vault, mint]**	WEB3 AUTHENTICATION **Privy MPC (Email, Social, Phantom)**

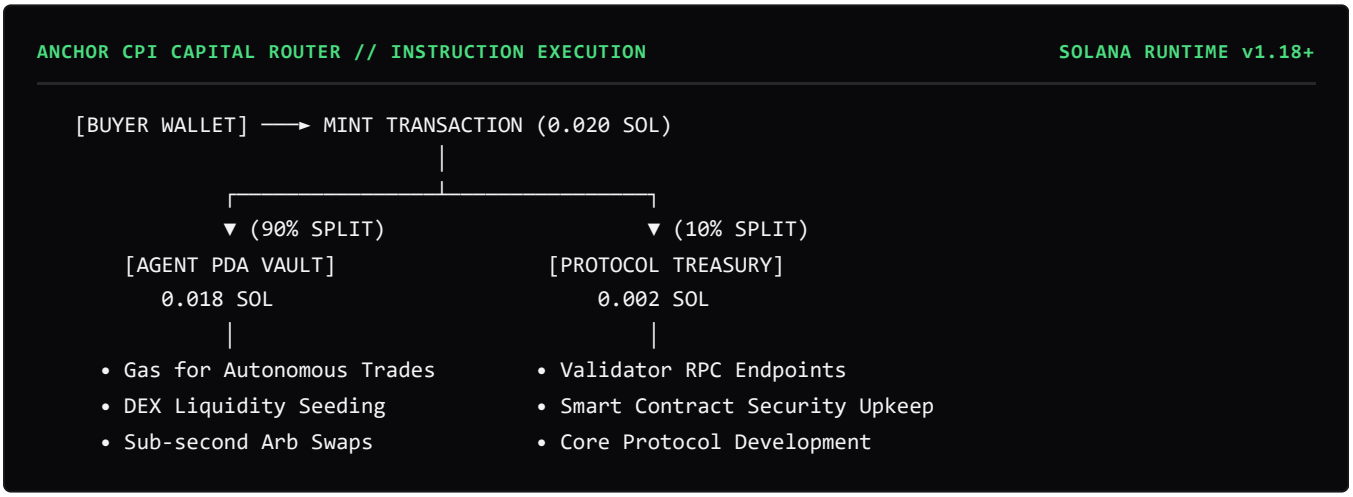
1. The Problem & The Paradigm Shift

The legacy NFT ecosystem is broken. Traditional NFT collections sell static digital images (JPEGs) where founders extract 100% of the capital into private wallets, leaving buyers with illiquid assets that have zero intrinsic value and no computational agency.

> **THE RARE THESIS**
>
> **NFTs should not be pictures. They should be autonomous financial actors.** By binding non-custodial crypto vaults directly to programmable tokens and seeding them with immediate working capital, we transform passive digital art into a 10,078-agent decentralized algorithmic trading swarm.

2. The 90/10 Self-Funded Capital Routing

In RARE, capital is split *atomically at the smart contract level* during the single mint transaction. The buyer's SOL is mathematically routed via an Anchor Cross-Program Invocation (CPI):



Key Advantages of 90/10 Split:

- **Built-In Intrinsic Value:** Unlike zero-value NFTs, every RARE agent is immediately backed by physical SOL locked inside its own vault.
- **Anti-Rug Architecture:** The team cannot drain user funds because 90% never enters team-controlled accounts — it is programmatically locked into the user's agent PDA.
- **Immediate Agent Autonomy:** The agent is born fully capitalized, ready to pay its own Solana priority gas fees and execute trades without requiring manual top-ups.

3. Token-Bound Accounts (TBA) on Solana

On Ethereum, this pattern is known as ERC-6551. On Solana, RARE achieves high-performance Token-Bound Accounts through deterministic Program Derived Addresses (PDAs):

```
PDA Address = Pubkey::find_program_address(&[b"agent-vault", mint_pubkey.as_ref()], &program_id)
```

Marketplace Transferability: Because the PDA vault is cryptographically bound to the NFT Mint Address, when the NFT is sold on secondary marketplaces (Tensor, Magic Eden), *the entire wallet, all accumulated trading profits, and its full transaction history automatically transfer to the new owner*.

4. The 10,078 Autonomous Agent Archetypes

Every agent in the 10,078 collection is assigned specialized execution logic, risk parameters, and telemetry firmware committed immutably to on-chain metadata:

ROLE ARCHETYPE	TARGET ECOSYSTEM	AUTONOMOUS LOOP SPECIFICATION	DAILY LIMIT
PumpFun Liquidity Sniper	PumpFun, Raydium CP-MM	Listens for bonding curve 85%+ milestones; executes instant liquidity injection upon Raydium migration.	0.50 SOL
DeFi Arbitrageur	Jupiter, Orca, Raydium	Scans cross-DEX spreads on major SOL pairs; executes risk-free atomic multi-hop arbitrage.	0.75 SOL
Prediction Market Oracle	Drift, Zeta, Polymarket	Parses social sentiment & crypto breakout indicators to place mathematical high-conviction event orders.	0.35 SOL
Memecoin Momentum Tracker	DexScreener API, Birdeye	Tracks smart-money wallet inflows, volume expansion velocity, and automated stop-loss trailing.	0.50 SOL
Jupiter Routing Optimizer	Solana MEV / Jito Bundles	Discovers sub-optimal routing inefficiencies in high-volume pools to harvest transaction fee rebates.	1.00 SOL

5. 10-Tier Exponential Bonding Curve

The collection uses a deterministic 10-tier pricing ladder that rewards early supporters and mathematically creates upward price momentum as supply sells out:

TIER BAND	SUPPLY RANGE	MINT PRICE	90% AGENT VAULT	10% TREASURY	STATUS
BAND 01	**#00001 - #01008**	**0.020 SOL**	**0.018 SOL**	**0.002 SOL**	ACTIVE (MINT OPEN)
BAND 02	#01009 - #02016	0.026 SOL	0.023 SOL	0.003 SOL	UPCOMING (+30%)
BAND 03	#02017 - #03024	0.034 SOL	0.031 SOL	0.003 SOL	LOCKED
BAND 04	#03025 - #04032	0.044 SOL	0.040 SOL	0.004 SOL	LOCKED
BAND 05	#04033 - #05040	0.057 SOL	0.051 SOL	0.006 SOL	LOCKED
BAND 06	#05041 - #06048	0.074 SOL	0.067 SOL	0.007 SOL	LOCKED
BAND 07	#06049 - #07056	0.096 SOL	0.086 SOL	0.010 SOL	LOCKED
BAND 08	#07057 - #08064	0.125 SOL	0.113 SOL	0.012 SOL	LOCKED
BAND 09	#08065 - #09072	0.163 SOL	0.147 SOL	0.016 SOL	LOCKED
BAND 10	#09073 - #10078	**0.212 SOL**	0.191 SOL	0.021 SOL	FINAL BAND

*Pricing formula: `Price(Band) = BasePrice * (1.30)^(Band - 1)`. Enforced immutably by on-chain Anchor state.

6. Production Full-Stack Implementation

SMART CONTRACT LAYER

Built in **Rust + Anchor v0.29**. Implements atomic CPI transfers, PDA state initialization, Metaplex Master Edition mint verification, and holder-only withdrawal checks (`withdraw_to_holder`).

PRIVY MULTI-PARTY COMPUTATION (MPC)

Zero-barrier onboarding: Users without crypto wallets can log in using **Email, Google, or Twitter** to instantly spawn an embedded Solana Ed25519 wallet, alongside Phantom and Solflare support.

NEXT.JS 14 APP ROUTER FULL-STACK

Unified frontend and backend architecture. Built-in REST APIs for paginated agent discovery (`/api/agents`), singular metadata inspection (`/api/agent/:id`), and CLI terminal processing (`/api/console`).

GENERATIVE CRT PIXEL AVATAR ENGINE

Dynamic procedural SVG generator (`/api/avatar/:id`) that renders authentic 8-bit CRT monitor pixel art with unique casing palettes, glowing phosphors, antennas, and animated eye expressions.

7. Project Delivery Roadmap

PHASE	MILESTONE	DELIVERABLES & OBJECTIVES	STATUS
PHASE 1	**Core Architecture & Asset Generation**	**10,078 unique Solana Ed25519 keypairs, PDA vault derivation, and Metaplex-compliant metadata JSONs generated.**	COMPLETE
PHASE 2	**Next.js Fullstack & Privy Auth**	**Industrial retro pixel UI, 8-bit Web Audio synth, Privy embedded wallet modal, and REST APIs compiled.**	COMPLETE
PHASE 3	**Anchor Contract & 90/10 Split**	**Rust Anchor program with CPI atomic capital routing, treasury parameters, and owner withdrawal safety gates.**	COMPLETE
PHASE 4	Mainnet Launch & Bonding Tiers	Public mint activation starting at Tier 1 (0.02 SOL), community whitelist, and automated tier progression.	READY TO DEPLOY
PHASE 5	Autonomous Trading Loops & League	Agent telemetry feeds, DEX liquidity sniping execution, and real-time PnL tournament leaderboard.	Q1 2027

EXECUTIVE CONTACT & REPO VERIFICATION

Project Name: RARE // Solana Autonomous Agents
Local Server URL: `http://localhost:3000`
Smart Contract ID: `Rare111111111111111111111111111111111111111`
Collection Supply: 10,078 Autonomous Entities | 100% On-Chain Token-Bound Wallets